



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

End-Autumn Semester Examination 2024-25

Date of Examination: 22-11-2024

Session: (FN/AN) FN Duration: 3 Hrs

Full Marks: 100

Subject No.: MT30001

Subject: Materials Engineering

Department/Center/School: Metallurgical and Materials Engineering

Specific charts, graph paper, log book etc., required: No

Special Instructions (if any): Answer All the Questions

Write the answers to Section I and Section II on separate sheets mentioning the Section names

SECTION-I

[(5x6)+(2x8)+10+(7x2)=70]

1. Find out the interplanar or d-spacing of (100), (110), and (111) crystallographic planes in a simple cubic system in terms of lattice parameter, a . [6]
2. Show with proper schematics and reasoning the number of symmetry elements present in a cubic crystal. [6]
3. Explain with proper reasoning whether the following sentence is true or false. [6]
The modulus of elasticity for single-crystal Iron is the same along $\langle 111 \rangle$, $\langle 110 \rangle$, and $\langle 100 \rangle$.
4. What are the slip planes and slip directions present in the body-centred cubic (BCC) material? How many of them are present in it? [6]
5. How many slip systems are required for homogenous plastic deformation of polycrystalline material? Explain your answer with proper reasoning. [6]
6. Explain how constitutional supercooling occurs in Cu-25wt%Zn single phase binary brass alloy with proper schematics. [8]
7. Discuss the different stages of annealing of a cold-worked polycrystalline material. How does it affect the microstructure? [8]
8. With the help of the Fe-Fe₃C equilibrium phase diagram, explain the microstructure evolution during the cooling of hypoeutectoid, eutectoid, and hypereutectoid steel under equilibrium conditions from the temperature above the Ac₃ line. [10]
9. A piece of Aluminium alloy with initial gauge length of 250 mm is pulled in tension up to the yield strength of 240 MPa. What would be the (i) True-stress at yield, (ii) elastic engineering strain, and (iii) true strain considering $E=110\text{GPa}$? If the uniform plastic engineering strain is 0.3, what would be the value of (iv) ultimate tensile strength, (v) uniform true stress, (vi) uniform true strain, and (vii) the strain hardening exponent, n , considering the *Ludwik* equation with the strength coefficient, $K=300\text{ MPa}$? [14]

SECTION-II

[(4x2)+(3x2)+6+(2x3)+4 = 30]

"The answers to this section should be on a separate sheet with SECTION-II as heading,"

1. A component is subjected to cyclic loading with maximum tensile stress of 500 MPa and minimum compressive stress of 300 MPa. Calculate the cyclic stress range, stress ratio, stress amplitude and mean stress. [8]
2. Determine the composition, in atom percent of an alloy that consists of 60 wt% sodium and 5 wt% chlorine. Assume the atomic weight of sodium is 23 and that of chlorine is 35.5. [6]
3. A plate of iron is exposed to a carburizing (carbon-rich) atmosphere on one side and a decarburizing (carbon-deficient) atmosphere on the other side at 500 °C. If a condition of steady state is achieved, calculate the diffusion flux of carbon through the plate if the concentrations of carbon at positions of 4 and 8 mm beneath the carburizing surface are 2.2 and 0.6 kg/m³, respectively. Assume a diffusion coefficient of $4 \times 10^{-11}\text{ m}^2/\text{s}$ at this temperature. [6]
4. Write the electronic configuration for Manganese, Nickel and Titanium. [6]
5. State Hume-Rothery rules. [4]